Ernie Chu (朱劭璿)

ML Undergraduate Researcher · Full-Stack Web Developer

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Education

National Sun Yat-sen University (NSYSU) — *Kaohsiung, Taiwan, R.O.C*

B.S. in Computer Science and Engineering — *Sep. 2018 - Expected Jun. 2022*

- GPA 4.15/4.3 (first three years)
- Two times dean's list award (書香獎) receiver
- MOST Research Grant for Undergraduate Students 獲科技部大專學生研究計畫補助
- **Second Place** in the Special Topic Study Contest for the Undergraduate Programs 2021 專題成果發表展第二名

Skills

General	C, Modern C++(17), Python, Java, Unix, LaTeX, Electron
Machine Learning	PyTorch, TensorFlow
Web Development	Vue.js, Express.js, Flask, PHP, HTML, CSS, JavaScript
Multimedia	WebGL, Final Cut Pro, PhotoShop
Languages	Chinese (native), English (TOEFL iBT 101), French

Experiences

Office of International Affairs, NSYSU — *Kaohsiung, Taiwan, R.O.C*

Full-Stack Web Developer (Part-Time) — *Jun. 2019 - PRESENT*

- Maintain the application system for exchange program using PHP, Nginx and MariaDB.
- Design and build the exchange program platform from scratch using Vue.js, Express.js, Sequelize.js and MariaDB.
- Maintain the official website for each division incorporating with Rpage, HeimaVista, Inc.

Northwest Analytics Inc - Taiwan — *Kaohsiung, Taiwan, R.O.C*

Data Engineer Intern — *Jun. 2018 - Jun. 2019*

- Data preprocessing for factory monitoring.
- Foundational Industry 4.0 Analytics.

Honors & Awards

2020	**Second Preliminary Round 初賽第二輪**, ATCC 18th ATONA Case Competition, ATCC 全國大專院校商業個案大賽	*Taiwan, R.O.C*
2020	**Excellent Achievement (Top 10) 佳作**, AICUP 2020 5 Types of Defects in Irwin Mango Classification, AICUP 全國大專校院人工智慧競賽愛文芒果五類不良品辨識競賽	*Taiwan, R.O.C*

Projects & Researches

TSM-Net: Temporal Compressing Autoencoder with Adversarial Losses for Time-Scale Modification on Audio Signals — *NSYSU, Taiwan, R.O.C*

Developer, Researcher & Co-Author — *Jan. 2021 - Oct. 2021*

- Use an autoencoder to compress the audio into 1024 times smaller latent representation for time-scale modification.
- Use multi-scale discriminator to train the model for the best audio quality on all frequency bands.
- **Paper**: https://ernestchu.github.io/tsm-net/tsm-net.pdf
- **Poster**: https://ernestchu.github.io/tsm-net-poster/poster.pdf
- **Talk**: https://youtu.be/h69v7_ZOKCM
- **Demo**: https://ernestchu.github.io/tsm-net-demo/

NSYSU Captcha Solver

NSYSU, Taiwan, R.O.C

Leading Developer — *Jan. 2021 - Feb. 2021*

- Use convolutional neural networks to autofill the captcha verification in the course inquiry system.
- Use TensorFlow.js in Chrome extension to implement on-device inference.
- **Repository**: https://github.com/ernestchu/nsysu-captcha-solver.

Secure On-line Chatting Service

NSYSU, Taiwan, R.O.C

Developer & Co-Author — *Spring 2021*

- On-line chatting service with console-based UI using ncurses.
- Socket programming and multi-threading using C++17
- Use attribute-based encryption to provide secure channel for messages.
- **Repository**: https://github.com/ernestchu/on-line-chatting-service/tree/abe.
- **Demonstartion**: https://youtu.be/S-j5kQz_-7Y.
- **Paper**: https://ernestchu.github.io/abe-examples/paper.pdf.

License Plate Verify and Warning System

NSYSU, Taiwan, R.O.C

Leading Developer — *Fall 2020*

- Use ML-based object detection technology to track the license plate.
- The barrier gate is no longer needed as the system can detect moving target.
- All of the inferences can be done on end-device, for example Jetson Nano.
- **Repository**: https://github.com/ernestchu/license-plate-verify-warning-system.
- **Demonstration**: https://youtu.be/jDKGAp2Puh0.

Deep Generative Models For ECG Denoising And Filtering

NSYSU, Taiwan, R.O.C

Developer, Researcher & Co-Author — *Fall 2020*

- Utilize domain-transferring generative models to mitigate the noise within ECG signals.
- This work deals with the classification of realistic noisy data and provides pragmatic results.
- **Paper**: https://ernestchu.github.io/noise-ecg-classification/paper.pdf.

Writings

CSE350 Computer Network Course Notes

NSYSU, Taiwan, R.O.C

Writer — *Spring 2021*

- https://ernestchu.github.io/courses/cse350-computer-network/

CSE365 Unix System Programming Course Notes

NSYSU, Taiwan, R.O.C

Writer — *Spring 2021*

- https://ernestchu.github.io/courses/cse365-unix-system-programming/

CSE360 Design And Implementation of Compiler Course Notes

NSYSU, Taiwan, R.O.C

Writer — *Spring 2021*

- https://ernestchu.github.io/courses/cse360-design-and-implementation-of-compiler/

CSE491 Network Application Programming Course Notes

NSYSU, Taiwan, R.O.C

Writer — *Spring 2021*

- https://ernestchu.github.io/courses/cse491-network-application-programming/